MANCHESTER 1824

The University of Manchester

EART11200 - The Natural Scientist's Toolkit

Mathematics and Physics

Week 3 Practical - Differentiating Combinations of Functions (Solutions)

Course layout

Date	Week	Topic
27/01/2025	20	Calculating Gradients
03/02/2025	21	Introduction to Differentiation
10/02/2025	22	Differentiating Combinations of Functions
17/02/2025	23	Optimisation & Properties of Derivatives
24/02/2025	24	Second Order Derivatives & Root Finding
03/03/2025	25	Introduction to Integration
10/03/2025	26	Integration using Newtons Laws
17/03/2025	27	Integration by Substitution
24/03/2025	28	Integrating Second Order ODEs
31/03/2025	29	Vectors and Partial Differentiation
07/04/2025	30	Non-teaching week
14/04/2025	31	Easter break
21/04/2025	32	Easter break
28/04/2025	33	Revision: Differentiation
05/05/2025	34	Revision: Integration
Exam Period		

Intended Learning Outcomes

(1) Apply the chain rule to differentiate composite functions in theoretical and real-world applications.

(2) Apply the product rule to differentiate products of functions in theoretical and real-world applications.

(3) Apply the quotient rule to differentiate products of functions in theoretical and real-world applications.

Exercise 1: Chain rule

- Use the chain rule to differentiate the following function:

$$f(x) = -\ln(\cos(x))$$

The Chain Rule

$$\frac{df}{dx} = \frac{df}{dg} \times \frac{dg}{dx}$$

Function $y = f(x)$	Derivative $\frac{dy}{dx}$
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

Solution 1: Chain rule

The Chain Rule

$$\frac{df}{dx} = \frac{df}{dg} \times \frac{dg}{dx}$$

- **Step 1**: Split the function and substitute

$$f(g) = -\ln(g) \qquad g(x) = \cos(x)$$

- **Step 2**: Differentiate:

$$\frac{df}{dg} = -\frac{1}{g} \qquad \frac{dg}{dx} = -\sin(x)$$

- **Step 3**: Apply the product rule and substitute back in:

$$\frac{df}{dx} = \frac{\sin(x)}{\cos(x)} = \tan(x)$$

Function $y = f(x)$	Derivative $\frac{dy}{dx}$
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

Exercise 2: Product rule

- Use the product rule to differentiate the following function:

$$g(x) = \sin(x)\exp(x)$$

The Product Rule

$$\frac{df}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

Function $y = f(x)$	Derivative $\frac{dy}{dx}$
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

Solution 2: Product rule

- **Step 1**: Split the function and substitute

$$u(x) = \sin(x) \qquad v(x) = \exp(x)$$

- **Step 2**: Differentiate:

$$\frac{du}{dx} = \cos(x) \qquad \frac{dv}{dx} = \exp(x)$$

- **Step 3**: Apply the product rule and substitute back in:

$$\begin{aligned}\frac{dg}{dx} &= \cos(x)\exp(x) + \exp(x)\sin(x) \\ &= \exp(x)(\cos(x) + \sin(x))\end{aligned}$$

The Product Rule

$$\frac{df}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

Function $y = f(x)$	Derivative $\frac{dy}{dx}$
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

Exercise 3: Quotient rule

- Use the quotient rule to differentiate the following function:

$$h(x) = \frac{x - x^2}{1 + 8x^2}$$

The Quotient Rule

$$\frac{df}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Function $y = f(x)$	Derivative $\frac{dy}{dx}$
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

Solution 3: Quotient rule

- **Step 1**: Split the function:

$$u(x) = x - x^2 \qquad v(x) = 1 + 8x^2$$

- **Step 2**: Differentiate:

$$\frac{du}{dx} = 1 - 2x \qquad \frac{dv}{dx} = 16x$$

- **Step 3**: Apply the product rule and substitute back in:

$$\begin{aligned}\frac{dh}{dx} &= \frac{(1+8x^2)(1-2x) - 16x(x-x^2)}{(1+8x^2)^2} \\ &= \frac{1 - 2x + 8x^2 - 16x^3 - 16x^3 + 16x^3}{(1+8x^2)^2} \\ &= \frac{1 - 2x - 8x^2}{(1+8x^2)^2}\end{aligned}$$

The Quotient Rule

$$\frac{df}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Function $y = f(x)$	Derivative $\frac{dy}{dx}$
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

Properties of Derivatives

- To finish up, we can use derivatives to tell us something about the function.
- We know that if

$$\frac{dy}{dx} > 0 \implies \text{function is increasing}$$
$$\frac{dy}{dx} < 0 \implies \text{function is decreasing}$$

- We can then deduce the if

$$\frac{dy}{dx} = 0$$

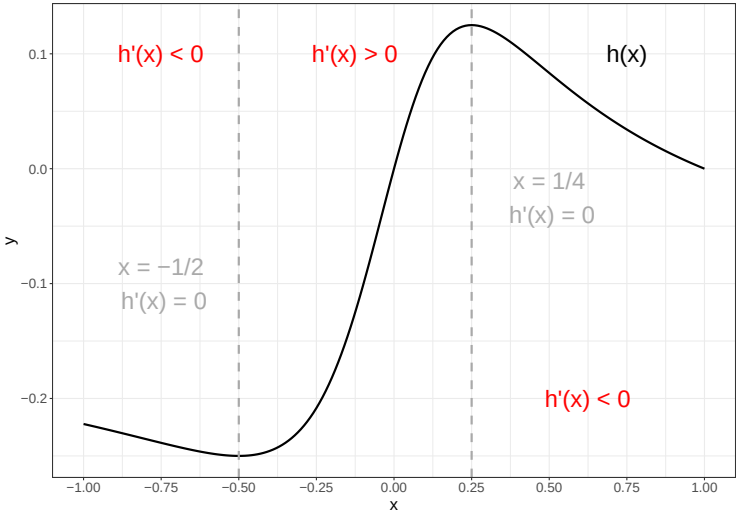
  then the function must be changing direction.
- We will come across the important property a zero derivative has soon.

Properties of Derivatives

- For the function:

$$h(x) = \frac{x - x^2}{1 + 8x^2}$$

$$\frac{dh}{dx} = \frac{1 - 2x - 8x^2}{(1 + 8x^2)^2}$$



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